

Advancements in FPGAs and Development Boards Look to Accelerate Time-To-Market



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To evaluate a device and develop an application, field programmable gate array (FPGA) based development or evaluation boards are a great platform. From engineers and hobbyists who want to start with FPGAs to professionals looking for mid- or high-end boards for their prototyping and designing needs, there is a plethora of products out there.

In today's market, the changing system-level requirements crop the need for device-level changes. Hence, the addition

of new features to evaluation and development boards have become a necessity, feels Sachin Gupta, product marketing engineer, SoC product-line marketing, Microsemi. He says, "Given the current scenario, where cyber weapons are trying to break into embedded designs more aggressively, the challenges set before us to countermeasure and protect data and designs from these attacks are higher than ever."

He adds, "Other traditional security concerns are protection of design from

Latest FPGA Developments Boards, SoC Kits and Modules in the Market

Company	Product	Features
Avnet	Artix-7 50T FPGA evaluation kit 	- Customisable development kit perfect for embedded designers looking for a flexible, low-power platform - Packages features like 256MB DDR3 SDRAM, 32MB of QSPI flash, 32KB of I2C EEPROM, dual 10/100 Ethernet interfaces, six Digilent compatible Pmod interfaces enabling 48 user I/O pins - Solution for designers interested in exploring MicroBlaze soft processor or Artix-7 FPGAs in general
BittWare	A10PL4 	- Arria 10 GT/GX low-profile PCIe board with dual QSFP and DDR4 - Features Altera Arria 10 GT/GX FPGA, PCIe x8 interface supporting Gen1, Gen2 or Gen3, dual QSFP cages for 2x 100GigE, 2x 40GigE, or 8x 10GigE and memory up to 32GB of DDR4 SDRAM with ECC (x72) - Configuration via protocol (CvP) supported
BittWare	Block diagram A10P3S 	- Arria 10 GT/GX/SX 3/4-length PCIe board with quad QSFP DDR4 and QDR-IV - Features Altera Arria 10 GT/GX/SX FPGA and SoC, PCIe x8 interface supporting Gen1, Gen2 or Gen3, four QSFP cages for 4x 100GigE, 4x 40GigE, or 16x 10GigE and board management controller for intelligent platform management
Embest	Lark board 	- Based on Altera's Cyclone V SoC, Lark Board is designed for development of high-volume applications including automotive, medical equipment, video surveillance and industrial control - Features a 5CSXFC6D6 SoC FPGA, 1GB DDR3 SDRAM to each of the HPS and FPGA as well as 4GB eMMC flash, Transflash card slot for mass storage, several standard I/O interfaces including PCI Express, Gigabit Ethernet, USB2.0 Host ports, UART, JTAG and 3G SDI
Pro Design	Zynq XC7Z045 	- Combines a user FPGA with an ARM core processor (dual ARM Cortex-A9 MPCore with CoreSight) and several onboard peripherals such as USB 2.0 OTG, gigabit Ethernet or ARM JTAG debug - Includes 1GB onboard DDR3 and dual quad-SPI flash memory - Allows maximum point-to-point speed of up to 1.2Gbit/s over the standard FPGA I/O and up to 12.5Gbit/s over high-speed serial transceivers
Scarab Hardware	miniSpartan6+ 	- Features Spartan6 LX9 Board from Xilinx, an onboard USB JTAG programmer to power and program the FPGA with any open source programmer, like the one inside Scarab IDE - Also includes an onboard USB interface that powers the board and allows communication with the PC at speeds up to 480Mbps, HDMI port, 8-channel analogue-to-digital converter running at 1MSPS with 8-bit resolution and memory of 32MB of SDRAM, 64Mbit of SPI flash and a microSD card interface
Terasic	SoCKit development kit 	- Built around Altera Cyclone V SoC FPGA, which combines the latest Cortex-A9 embedded cores - Includes hardware such as high-speed DDR3 memory, video and audio capabilities, Ethernet networking and an onboard HSMC connector with high-speed transceivers, which allows for an even greater array of hardware setups